Prakhar Gandhi — (+91)-9133234325 | gprakhar0@gmail.com | ResearchGate
Github | Portfolio | Website | Utube | Multimodal ML Pipeline | Automation | ChatBot | MapReduce | Scalable Image Features App | Webapp | Shor's Algorithm | Multiprocessing | AGI | Quantum ML | Universal Algorithms Visualizer | Scalable Distance Metrics | Scalable Folder Search | Scalable OCR | Software Dev | Agentic AI | Automate Imputation | Optimized Dashboard Automation | Optimized Content Generation

Summary

Lead AI Engineer with 6+ years of experience in Python, ML, and Generative AI. Proven record of building scalable ML pipelines, agentic AI systems, and internal products used by 1000+ users. Strong background in NLP, computer vision, and parallel computing with measurable performance gains. Experienced in mentoring and cross-team leadership.

Education

BITS Pilani - B.E. in Electronics and Instrumentation — Hyderabad, Telangana
Jan. 2015 – Feb 2020

Academic & Professional Development

Coursework: Data Structures and Algorithms, Neural Networks and Fuzzy Logic, Discrete Math for Computer Science
Self Taught Courses: Database Management Systems, Number Theory: Project Euler
Certifications: Deep Learning: Advanced NLP and CNNs, RNNs, LSTMs and GRUs, Bayesian Machine Learning in Python: A/B Testing, Recommender Systems and Deep Learning in Python, CNN in python - Computer Vision
Certificates Link
Awards: Have received a gold award from Standard Chartered Bank Global Business Services Private Limited as a part of AI/ML Hackathon, Victory League Award in Ericsson

Technical Skills

Programming Languages: Python, C++, JavaScript
Machine Learning & AI: NLP, Computer Vision, Generative AI, RAG, Model Evaluation, Feature Engineering
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, Langgraph, Crew AI
Backend & APIs: Dash, Django, Fastapi, Microservices, Tool calling functions for CRUD
Data & vector databases: Qdrant, Snowflake, Testing & Documentation Automation (Playwright, pdoc)
Cloud & DevOps: Docker, Terraform, Azure Functions, Agentic AI using Azure Foundry, CI/CD, Amazon Q, Github Copilot.

Experience

Technical Lead Architect (AIML Computational Specialist - Generative AI) — April 2026 – Present
Accenture - • AT&T • Hunter Douglass Solution • Campbell • Linde Voyager • Coty • Diebold — *Bangalore, IN*

- Scaled up the contract handling using guardrail mechanism/custom field validation using fuzzylogic+easyocr.
- Designed a production-grade CRUD pipeline integrated with Snowflake for contract extraction, reducing manual processing effort.

Senior AI Engineer (Python Developer - Generative AI) — Feb 2025 – April 2026
Wipro - Ericsson India Private Limited — *Bangalore, IN*

- Developed a ChatGPT-style user interface with CRUD features in Dash + SQLite
- Demo presented to 1000+ Ericsson employees.
- Quality Features included Pagination, Dark/Light Theme Toggle, Export to CSV support for tables, Interactive Zoomable/Drillable Graphs (Plotly), Textual/Tabular/Graph formatting.
- Implemented Agentic AI workflows using MCP, LLM-Based Evaluation, RAG, pydantic, chunking and retrieval, Reinforcement Learning Human Feedback, recursive text splitter, reranker mechanism, langchain, openai-models(GPT-4/3.5, Groq), Planner Agents, Graph construction, Orchestration Agent.

Senior Software Engineer (Python Developer) — May 2024 – Jan 2025
Wipro - Mercedes-Benz Research and Development India — *Bangalore, IN*

- Automated manual cpp file code generation by doing Parameter and function name extraction of car product cpp files using python and multiprocessing. Optimized 2nd level folder search under 2 mins using parallel processing.
- Developed a Flask application that includes live terminal using socketio and docker integration in end to end GUI.

Data Analyst 2 Aug 2020 – Jan 2024
Standard Chartered Bank Global Business Services Private Limited India *Bangalore, IN*

- Automated Backward Elimination ML procedure for correlated features removal.
- Developed a Tensorspace visualization custom component to solve the memory leakage issues at hackathon.
- Developed a flask app to visualize model predictions and flipped classes based on synonym replacement, insertion and deletion of tense words of banking words present in a general spacy dictionary to check the robustness of model to any such attacks.
- Implemented multiprocessing on django microservices by using parallel processing using pandarallel package and decreased the execution time.
- Developed a QA generation pipeline from scratch using the knowledge of competitive programming by invoking the Large Language Models named t5-base-squad of medium parameter size. Speedup the entire process by using parallel processing.
- Automated synthetic dataset generation by doing Multi blending labels of product documents(watermarks, barcodes, seals and other confidential details). This work involved Parallel Processing improved Data Creation for the agreed deliverables from 4 hour to 1 hour per document.
- Implemented McNemar Test via statistical analysis for cross-team members from structured data to compare the performance of 2 classifiers and selecting the best model further integrating it within the champion challenger pipeline.

Projects - Standard Chartered

NER Explainability Creation | *Python, sklearn, pandas, numpy, plotly, dash* SCB

- Built local and global NER explainability visualizations including entity probability, POS importance, and state transition analysis.
- Implemented an end-to-end NLP pipeline for an ML-based NER explainer search engine.
- Deployed CRFsuite NER explainability using Dash, enabling accurate class-wise analysis at sentence level.
- Designed a scalable Champion–Challenger framework across NER, regression, classification, and object detection pipelines.
- Optimized evaluation metrics and system performance to support pipelines at million-scale data volume.
- Presented model explainability and performance insights to senior leadership.

Image duplicate prediction (Banking) | *Python, pandas, numpy, tensorflow(imagededup), pytorch(detecto), opencv*SCB

- Built ML-based duplicate detection systems for banking PDFs, improving identification of barcodes, seals, and key document artifacts by 80%.
- Developed object detection models for barcodes, watermarks, and signatures using Faster R-CNN with SSIM-enhanced similarity scoring, achieving 85% accuracy.
- Generated analytical visualizations and client-ready reports to communicate model performance and validation results.

Experience with Assignments (External)

Zycus assignment | *Python, pyspark, ML* Zycus

- Built a PySpark text classification pipeline for product reviews, achieving a 91% speedup **Github Link**

Happymonk paper implementation | *Python, pandas, numpy* Happymonk

- created a 3 layer artificial neural network from scratch as a part of research paper provided to me in coding assignment and gained 96% accuracy for benchmarks like iris/mnist dataset, **Github Link**

Achievements

Competitive Programming Ranks | *Python, Cpp* Contests

- Red Coder on Atcoder. **Link**

System Design, Cloud Architecture, Distributed Systems, AI Architecture, Leadership At Companies

- Mentored a senior colleague in Standard Chartered as a part of hackathon in visualizing robustness check of ML/DL model at a targetted classwise image level attack and made a opacity test to compare a permutation of two classes at a time by implementing the same from the paper. Further Detected poison in image dataset.

Hall of Famer! | *Python*

- Picasso Themed Scalable Blockchain Webapp without use of database to keep the randomization of nft images intact with encryption algorithm to keep passwords secure, added buttons and scaled it up to billion fold world,
  1) Github Link 2) Python Package if you prefer to run in local 3) Deployed Demo

- Implemented Love Hate Algorithm with face detection to find best colleague to work with and scaled it up,
  1) Github Link 2) Python Package if you prefer to run in local